

Autodesk® Scaleform®

Scaleform 4

Scaleform (Scaleform 3) Scaleform 4.0

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Autodesk® Scaleform® 4.4

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1 Scaleform 4 1

2 2

 2.1 2

 2.2 Scaleform 3

3

1 Scaleform 4

Scaleform v.4.1.44 Autodesk
 SDK , ActionScript 3.0 SDK ,Scaleform UI
 AS 2.0 AS 3.0 ,Scaleform v.4.1.44

- AS3, SDK ,Scaleform v.4.1.44
- w/ 2x+ – 2.5D
- - Scaleform 4 Advance Display
- – ,AS3 AS2 Scaleform 4
- - AS3 (Code Base), ,
- / – ,AS3
- – AS 2.0 , AS 3.0
 (Runtime Exception)
- – AS 2.0 , AS 3.0 ,
- – ActionScript 3.0 ,
 (Method Closure)

Scaleform 4 , Scaleform 4
 Scaleform 4.0 4.2 ,Scaleform 4 Scaleform 3.x
 , Scaleform
 Scaleform 4 , Scaleform C++
 : (NameSpace) ,

2

2.1

```
namespace
namespace Scaleform
{
    class Foo
    {
        ...
    };
}
```

J

```
1)      :      ,
      Scaleform::Foo var;
```

```
2)      'using'
        using namespace Scaleform;
        Foo var;
```

C++

Scaleform , "Scaleform"

- Scaleform , , "using"

- Scaleform "SF" (SF = Scaleform)
(Header) ; , , ,
- SF , : "Render" "Sound" "GFX"
- , , "Win32" "PS3",
, "Scaleform::Render::D3D9" , "SF::D3D9::Render"

2.2 Scaleform

Scaleform , Scaleform
(Public)

Scaleform Scaleform , , ,
Scaleform

Scaleform 4

- Scaleform
 - GFX
 - AMP
 - XML
 - Render
 - Math2D
 - D3D9
 - GL
 - PS3
 - ...
 - Text
 - Sound
 - Alg
 - UTF8Util

3

Scaleform

, Scaleform

,

,

,

- "Scaleform" Src
- "Src/Kernel"

- Src
 - Gfx
 - AMP
 - AS2
 - AS3
 - ...
 - Kernel
 - HeapMH
 - HeapPT
 - Render
 - PS3
 - GL
 - D3D9
 - ...
 - Sound
 - XML

Src

,

cpp

(Public)

,

4 Includes

Scaleform 3.x ,Scaleform 4 /Src , /Include
 .cpp

Scaleform ;Gfx_Kernel.h Kernel , '(convenience) ,Gfx.h
 , Include

,
 , include

```
#include "Kernel/SF_File.h"
#include "Gfx/Gfx_Player.h"
#include "Gfx/Gfx_Loader.h"
#include "Gfx/Gfx_Log.h"
```

```
#include "Gfx_Kernel.h"
#include "Gfx.h"
```

, ,

"PublicHeader : GroupName"

,GroupName , Gfx Kernel Render AMP ... none GroupName() Gfx,
 Gfx.h , GroupName Kernel, Gfx_Kernel.h ,
 GroupName 'none', - ,
 ,

5

Scaleform 4 API , , Scaleform , Scaleform (Scaleform 3.x SDK
), Scaleform 4 API
 Scaleform include Scaleform 4
 Scaleform 3.x Scaleform 4 Scaleform
 Scaleform Scaleform

4 (Getting Started with Scaleform 4.2)

Scaleform Scaleform 4
 Scaleform 4
 Scaleform

Scaleform 3.x
 Scaleform 4
 Scaleform 4 Scaleform
 Scaleform 4 (Scaleform 4Renderer Threading Guide)
 ,Scaleform 4 (Include/GFxCompat.h),
 Scaleform 3.x Scaleform 4 typedefs enums defines Scaleform 3.X
 Scaleform 4
 Scaleform 4
Apps/Demos/GFxPlayerTiny/GFxPlayerTinyD3D9Compat.cpp

GfxCompat.h Scaleform 4 ,
 , Gfx Loader Scaleform 4 :FontProvider AS2Support /
 AS3Support ,

```
// Set Font Provider:
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32 (::GetDC(0));
loader.SetFontProvider(fontProvider);

// Add AS2 Support:
Ptr<ASSupport> pAS2Support = *new GFx::AS2Support();
loader.SetAS2Support(pAS2Support);

// Add AS3 Support:
Ptr<ASSupport> pAS3Support = *new GFx::AS3Support();
```

```
loader.SetAS3Support(pASSupport);
```

```
,
Scaleform 3.x
GfxCompat.h
Scaleform 4
```

5.1

```
Scaleform 3.x
Scaleform 4
```

```
#include "Gfx_Kernel.h"
#include "Gfx.h"
#include "Gfx_Renderer_D3D9.h" // or whatever platform you need
```

```
(Gfx.h)
AS3
AS3
"Gfx/AS3/AS3_Global.h"
Scaleform LITE
AS3
```

AS3_Global.h and AS3_Obj_Global.xxx

```
:AS3_Global.h
ObjAS3_Obj_global.xxx
AS3_Obj_Global.xxx
"(script)
Scaleform VM
" "ActionScript 3
GlobalObjectCPP,
swf
C++
```

```
AS3_Global.h
ClassRegistrationTable
AS3 C++
```

```
ClassRegistrationTable,
```

```
AS3_Global.h
```

```
ClassRegistrationTable include
ClassRegistrationTable(
AS3 VM
" "
AS3_Global.h
```

5.2

, GFC SF_ Scaleform GFX_

Scaleform 3.X	Scaleform 4
GFC_FX_VERSION_STRING	GFX_VERSION_STRING
GUNUSED	SF_UNUSED
GFC_64BIT_POINTERS	SF_64BIT_POINTERS
...	...

5.3 Namespace

Scaleform 4 (Scaleform::Gfx::Event) using namespace' , using cpp ,

```
namespace SF = Scaleform;
using namespace Scaleform;
using namespace Render;
using namespace Gfx;
```

5.4

Scaleform 4 , ,

Scaleform 3.X	Scaleform 4
UInt	unsigned
SInt	int
Float	float
...	

Scaleform 4 Scaleform , ,
Scaleform 3.x , GfxEvent , Scaleform 4 , Gfx::Event
Scaleform 3.x Scaleform 4

Scaleform 3.X	Scaleform 4	Scaleform 4
		using namespace ;
GfxSystem	Scaleform::Gfx::System	System

GPtr	Scaleform::Ptr	Ptr
GFxMovieDef	Scaleform::GFx::MovieDef	MovieDef
GFxMovieView	Scaleform::GFx::Movie	Movie
GRendererD3D9	Scaleform::Render::D3D9::HAL	Render::D3D9::HAL
GString	Scaleform::String	String
GFxFSCCommandHandler	Scaleform::GFx::FSCCommandHandler	FSCCommandHandler
GFxLog	Scaleform::GFx::Log	Log
GFxImageCreator	Scaleform::GFx::ImageCreator	ImageCreator
GFxKey	Scaleform::GFx::Key	Key
GFxKeyEvent	Scaleform::GFx::KeyEvent	KeyEvent
GFxCharEvent	Scaleform::GFx::CharEvent	CharEvent
GFxEvent	Scaleform::GFx::Event	Event
GMatrix2D	Scaleform::Render::Matrix2x4	Matrix2x4
GMatrix3D	Scaleform::Render::Matrix3x4	Matrix3x4
GMatrix3D	Scaleform::Render::Matrix4x4	Matrix4x4
GRect	Scaleform::Render::Rect	Rect
...	...	...

6 Scaleform 4.0 4.2

Scaleform 4.2 4.0 API Scaleform 4.1.x
API 4.0

6.1

SF 4.2 PCRE Perl (Perl Compatible Regular
Expressions, PCRE), AS3 3rdParty , Scaleform 4.2
Include/GFxConfig.h PCRE AS3, PCRE(
SF_ENABLE_PCRE), PCRE

6.2

Scaleform 4.2 , , Scaleform Scaleform
(Build System)
Render() Sound() SF 4.2

Render Scaleform
X360/D3D9/D3D1x , InitHAL()
• HALConfig_DynamicShaderCompile

Visual Studio , : Render , ShaderData.xml,
Custom Build Step

6.3

Scaleform 4.2 -Scaleform 4.2
40 SF 4.0 Scaleform 4.0

gfxlicense.txt sf_license.txt
gfx.lib
,sf_license.txt Scaleform AMP Scaleform

Scaleform IME sf_video_license.txt
sf_ime_license.txt

Scaleform